



2024

L20

BOOKLET NO.

360051

Forest Services
Physics

Time Allowed : Three Hours

Maximum Marks : 200

Medium : English

Type of Paper : Conventional

Question Paper Specific Instructions

Please read each of the following instructions carefully before attempting questions :

1. There are **EIGHT** questions divided in two Sections out of which **FIVE** are to be attempted.
2. Questions No. 1 and 5 are compulsory. Out of the remaining questions, **THREE** are to be attempted choosing at least **ONE** question from each Section.
3. The number of marks carried by a question/sub question is indicated against it.
4. Keep in mind the word limit indicated in the question if any.
5. Wherever option has been given, only the required number of responses in the serial order attempted shall be assessed. Unless struck off, attempt of a question shall be counted even if attempted partly. Excess responses shall not be assessed and shall be ignored.
6. Candidates are expected to answer all the sub-questions of a question together. If sub-question of a question is attempted elsewhere (after leaving a few page or after attempting another question) the later sub-question shall be overlooked.
7. Any page or portion of the page left blank in the Answer Booklet must be clearly struck off.
8. Unless otherwise mentioned, symbol and notation have their usual standard meanings. Assume suitable data, if necessary and indicate the same clearly.
9. Neat sketches may be drawn, wherever required.
10. The medium of answer should be mentioned on the answer book as claimed in the application and printed on admission card. The answers written in medium other than the authorized medium will not be assessed and no marks will be assigned to them.

Note – 1. Candidates will be allowed to use Scientific (Non-programmable type) calculators.

P.T.O.

SEAL

**SECTION – A****Q1.** Solve **any five** out of seven.**(8×5=40)**

- (a) What is linear momentum ? State the principle of conservation of linear momentum. Explain the principle using an example.
- (b) Define the term interference. Show that in Young's double slit experiment, bright and dark bands are equally spaced.
- (c) What qualifies a material to be called as a dielectric ? How do you define the dielectric constant of a given material ? What happens when such material is kept in an electric field ?
- (d) Discuss the following with neat diagrams,
 - i) Self-inductance.
 - ii) Mutual-inductance.
- (e) Define the following and show on an indicator diagram,
 - i) Isothermal process
 - ii) Adiabatic process
 - iii) Isobaric process
 - iv) Isochoric process.
- (f) What is a blackbody ? Describe the characteristics of blackbody radiation with the help of a neat diagram. Give examples of blackbody.
- (g) Describe the structure of a typical optical fibre. Differentiate between step-index and graded index fibre.

Q2. (a) Obtain the moment of inertia of a solid, uniform sphere about an axis through its center. Calculate the moment of inertia of a solid sphere of mass 10 kg and radius 0.5 m, rotating about an axis 0.2 m from the centre of the sphere. **15**

- (b) Discuss the Fraunhofer diffraction at a circular aperture and obtain the radius of Airy's disc. In Fraunhofer diffraction pattern due to a circular aperture, the screen is at a distance of 1.5 m from the convex lens. The aperture is illuminated by sodium light of wavelength $5893\text{\AA}$. If the diameter of the aperture is 0.2 mm, find the separation between the central disc and the first minimum. **15**

- (c) Write down the Poisson and Laplace equations in electrostatics. Use Laplace law to compute the potential outside a sphere of uniform charge density ' ρ ' at a distance ' r '. **10**
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- Q3.** (a) Obtain an expression for the current in a circuit where an inductor and a resistance are connected in series with a voltage source. What is meant by 'time constant' for such circuits ? A relay having an inductance of 4H and resistance 200 Ω operates at 2 mA. How much time does the relay take to operate if potential difference of 1V is applied ? **15**
- (b) State the Laws at thermodynamics. Show that it is impossible to construct a heat engine which operates on a cycle and produces no other effect than the transfer of heat from a source in order to produce an equal amount of work. **15**
- (c) What is ferromagnetism ? How is it different from paramagnetism ? Draw a hysteresis loop for soft iron and explain why is it used to make electromagnet cores. **10**
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- Q4.** (a) Describe the construction and working of a Diesel engine. What is the difference in its working with respect to a petrol engine ? **15**
- (b) What is meant by stationary waves ? For a stationary wave formed in a stretched string, obtain the following,-
- i) Equation for displacement
 - ii) Maximum amplitude
 - iii) Position of nodes
 - iv) Position of antinodes
 - v) Fundamental frequency. **15**
- (c) Describe the construction and working of a He-Ne Laser. **10**
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SECTION - B

- Q5.** Solve **any five** out of seven. **(8×5=40)**
- (a) Define wave-particle duality.
 - (b) Write the significance of the Stern-Gerlach experiment.
 - (c) Define Q-value of a nuclear reaction.
 - (d) Give the difference between intrinsic and extrinsic semiconductor.
 - (e) What is the role of diffusion in biology ?
 - (f) Write the significance of parity violation in beta decay.
 - (g) Define osmotic pressure.
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- Q6.** (a) Derive the density of states for a particle in a three-dimensional box. **15**
(b) Discuss the normal Zeeman effect and Zeeman shift. **15**
(c) How laws of thermodynamics obeyed in the context of living organisms ? **10**
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- Q7.** (a) Explain the shell model of the nucleus and its limitations. **15**
(b) Discuss the working of logic gates and their truth tables. **15**
(c) What are the key physico-chemical properties of biological fluids ? And how do they influence physiological functions ? **10**
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- Q8.** (a) Discuss the reflection and transmission of a particle encountering a potential step. **15**
(b) Explain the working of a p-n-p transistor. **15**
(c) What are the LS and jj couplings in two valence electron atom. **10**
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